

that is

$$mz' - ny' = \frac{1}{K} \left\{ \lambda(A, \dots | \xi', \eta', \zeta')^2 - \xi'(A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\},$$

and similarly

$$nx' - lz' = \frac{1}{K} \left\{ \mu(A, \dots | \xi', \eta', \zeta')^2 - \eta'(A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\},$$

$$ly' - mx' = \frac{1}{K} \left\{ \nu(A, \dots | \xi', \eta', \zeta')^2 - \zeta'(A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\};$$

and thence

$$\begin{aligned} (A, H, G | mz' - ny', \dots) \\ = \frac{1}{K} \left\{ (A, H, G | \lambda, \mu, \nu) \cdot (A, \dots | \xi', \eta', \zeta')^2 \right. \\ \left. - (A, H, G | \xi', \eta', \zeta') \cdot (A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\} \end{aligned}$$

with the like equations, writing H, B, F and G, F, C in the place of A, H, G successively: and we then have

$$(A, \dots | mz' - ny', \dots)^2 = \frac{1}{K} \left\{ (A, \dots | \xi', \eta', \zeta' | mz' - ny', \dots) \cdot (A, \dots | \xi', \eta', \zeta')^2 - (A, \dots | \xi', \eta', \zeta')^2 \cdot (A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\}.$$

But the foregoing values of $mz' - ny'$, $nx' - lz'$, $ly' - mx'$ give also

$$(A, \dots | \xi', \eta', \zeta' | mz' - ny', \dots) = \frac{1}{K} \left\{ (A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \cdot (A, \dots | \xi', \eta', \zeta')^2 - (A, \dots | \xi', \eta', \zeta')^2 \cdot (A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\} = 0;$$

so that

$$(A, \dots | mz' - ny', \dots)^2 = -\frac{1}{K} (A, \dots | \xi', \eta', \zeta')^2 \cdot (a, \dots | \lambda x - \mu \xi', \dots)^2.$$

Similarly,

$$(A, \dots | mz' - ny', \dots | yz' - y'z, \dots) = \frac{1}{K} \left\{ (A, \dots | \xi', \eta', \zeta' | yz' - y'z, \dots) \cdot (A, \dots | \xi', \eta', \zeta')^2 - (A, \dots | \xi', \eta', \zeta')^2 \cdot (A, \dots | \xi', \eta', \zeta' | \lambda, \mu, \nu) \right\}.$$

But

$$\begin{aligned} (A, \dots | \xi', \eta', \zeta' | yz' - y'z, \dots) &= (A\lambda + H\mu + G\nu)(yz' - y'z) \\ &\quad + (H\lambda + B\mu + F\nu)(zx' - z'x) \\ &\quad + (G\lambda + F\mu + C\nu)(xy' - x'y), \\ &= x[y'(G\lambda + F\mu + C\nu) - z'(H\lambda + B\mu + F\nu)] \\ &\quad + y[z'(A\lambda + H\mu + G\nu) - x'(G\lambda + F\mu + C\nu)] \\ &\quad + z[x'(H\lambda + B\mu + F\nu) - y'(A\lambda + H\mu + G\nu)], \end{aligned}$$

which, substituting for x', y', z' their values

$$(x', y', z') = \frac{1}{K} \begin{pmatrix} A, H, G \\ H, B, F \\ G, F, C \end{pmatrix} (\xi', \eta', \zeta'),$$

becomes

$$\begin{aligned} &= \frac{1}{K} \left\{ x[(BC - F^2)(\nu\eta' - \mu\zeta') + (FG - CH)(\lambda\zeta' - \nu\xi') + (HF - BG)(\mu\xi' - \lambda\eta')] \right. \\ &\quad + y[(FG - CH)(\nu\eta' - \mu\zeta') + (CA - G^2)(\lambda\zeta' - \nu\xi') + (GH - AF)(\mu\xi' - \lambda\eta')] \\ &\quad \left. + z[(HF - BG)(\nu\eta' - \mu\zeta') + (GH - AF)(\lambda\zeta' - \nu\xi') + (AB - H^2)(\mu\xi' - \lambda\eta')] \right\}, \end{aligned}$$

which is

$$= (a, \dots | x, y, z | \nu\eta' - \mu\zeta', \dots);$$

and by merely writing (ξ', η', ζ') in the place of (λ, μ, ν) , we have

$$(A, \dots | \xi', \eta', \zeta' | yz' - y'z, \dots) = 0;$$

so that we find

$$(A, \dots | mz' - ny', \dots | yz' - y'z, \dots) = \frac{1}{K} (A, \dots | \xi', \eta', \zeta')^2 \cdot (a, \dots | x, y, z | \nu\eta' - \mu\zeta', \lambda\zeta' - \nu\xi', \mu\xi' - \lambda\eta').$$

$$-\frac{1}{K}(A, \dots | \xi', \eta', \zeta')^2 \cdot (A, \dots | \lambda, \mu, \nu | \xi, \eta, \zeta) \pm \sqrt{-(A, \dots | \xi', \eta', \zeta')^2} \begin{vmatrix} \xi, \eta, \zeta \\ \xi', \eta', \zeta' \\ \lambda, \mu, \nu \end{vmatrix}.$$

21. And also

$$(A, \dots | \lambda, \mu, \nu)^2 = \text{Quotient by } (a, \dots | x', y', z')^2 \text{ of}$$

$$\text{IV. } \begin{cases} K(\xi x' + \eta y' + \zeta z')^2 \\ + \text{Quotient by } K(A, \dots | ny' - mz', \dots)^2 \text{ of Product} \\ [K\Phi \pm \sqrt{-K}(a, \dots | x', y', z')^2 (A, \dots | ny' - mz', \dots | \lambda, \mu, \nu)], \end{cases}$$

where

$$\Phi = \begin{vmatrix} \xi & \eta & \zeta \\ ax' + hy' + gz', & hx' + by' + fz', & gx' + fy' + cz' \\ ny' - mz', & lz' - nx', & mx' - ly' \end{vmatrix},$$

which may also be written

$$= (a, \dots | x', y', z' | l, m, n)(x'\xi + y'\eta + z'\zeta) - (a, \dots | x', y', z')^2 \cdot (l\xi + m\eta + n\zeta).$$

22. The geometrical signification is obvious. The formulae I. and II. each of them show that the equation

$$(a, \dots | x, y, z)^2 = 0$$

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of the conic may be written in the form

$$W^2 + \frac{1}{M}QR = 0,$$

where $Q = 0, R = 0$ are any two tangents of the conic, and $W = 0$ is the line joining the points of contact, or chord of contact corresponding to the two tangents; viz., in the formula I. we have

$$W = (a, \dots | x', y', z' | x, y, z),$$

$$\left. \begin{matrix} Q \\ R \end{matrix} \right\} = (A, \dots | mz' - ny', \dots | yz' - y'z, \dots) \pm \sqrt{-K(a, \dots | x, y, z)^2} \begin{vmatrix} x, y, z \\ x', y', z' \\ l, m, n \end{vmatrix},$$

(or for a different form of Q, R see the formula). The quantities (x', y', z') are the coordinates of the point of intersection of the two tangents, or pole of the chord of contact; (l, m, n) are supernumerary arbitrary quantities, the values whereof do not affect the result¹. And in the formula II. we have

$$W = \xi'x + \eta'y + \zeta'z,$$

$$\left. \begin{matrix} Q \\ R \end{matrix} \right\} = \Pi \pm \sqrt{-(A, \dots | \xi', \eta', \zeta')^2} (a, \dots | \nu\eta' - \mu\zeta', \dots | x, y, z),$$

(for the value of Π see the formula). The quantities (ξ', η', ζ') are the line-coordinates of the chord of contact (viz. the point-equation of this line is $\xi'x + \eta'y + \zeta'z = 0$); (λ, μ, ν) are supernumerary arbitrary quantities.

23. In the like manner the formulae III. and IV. each of them show that the line-equation

$$(A, \dots | \xi, \eta, \zeta)^2 = 0$$

of the conic may be written in the form

$$W^2 + \frac{1}{M}QR = 0,$$

¹In a different point of view, viz. if we consider the formula I. as a transformation of the function $(a, \dots | x, y, z)^2$, then (x', y', z') and (l, m, n) would be each of them supernumerary arbitrary quantities; and so in the other like cases.

where $Q = 0, R = 0$ are any two ineunts of the conic, and $W = 0$ is the point of intersection of the corresponding tangents; viz. in the formula III. we have

$$W = (a, \dots | \nu\eta' - \mu\zeta', \dots | \xi, \eta, \zeta),$$

$$\left. \begin{matrix} Q \\ R \end{matrix} \right\} = (a, \dots | \nu\eta' - \mu\zeta', \dots | \lambda\zeta' - \nu\xi', \dots) \pm \sqrt{-(A, \dots | \xi', \eta', \zeta')^2} \begin{vmatrix} \xi, \eta, \zeta \\ \xi', \eta', \zeta' \\ \lambda, \mu, \nu \end{vmatrix},$$

(for another form of Q, R see the formula).

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The quantities ξ', η', ζ' are the line-coordinates of the line through the two ineunts, or chord of contact; (λ, μ, ν) are supernumerary arbitrary quantities; and so in the formula IV. we have

$$W = x'\xi + y'\eta + z'\zeta,$$

$$\left. \begin{matrix} Q \\ R \end{matrix} \right\} = K\Phi \pm \sqrt{-K(a, \dots | x', y', z')^2} (A, \dots | ny' - mz', \dots | \xi, \eta, \zeta),$$

(for the value of Φ see the formula), where x', y', z' are the point-coordinates of the intersection of tangents at the two ineunts, or pole of the chord of contact; (l, m, n) are supernumerary arbitrary quantities.

24. We may, instead of the supernumerary arbitrary quantities (l, m, n) of the formula I., introduce the quantities (λ, μ, ν) , where

$$(l, m, n) = \frac{1}{K} \begin{pmatrix} A, H, G \\ H, B, F \\ G, F, C \end{pmatrix} (\lambda, \mu, \nu).$$

This gives

$$\begin{aligned} (A, H, G | mz' - ny', \dots) &= A(mz' - ny') + H(nx' - lz') + G(ly' - mx') \\ &= x'(Hn - Gm) + y'(Gl - An) + z'(Am - Hl) \\ &= \frac{1}{K} \left\{ x' [H(G\lambda + F\mu + C\nu) - G(H\lambda + B\mu + F\nu)] \right. \\ &\quad + y' [G(A\lambda + H\mu + G\nu) - A(G\lambda + F\mu + C\nu)] \\ &\quad \left. + z' [A(H\lambda + B\mu + F\nu) - H(A\lambda + H\mu + G\nu)] \right\} \\ &= x'(g\mu - h\nu) + y'(f\mu - b\nu) + z'(c\mu - f\nu) \\ &= \mu(gx' + fy' + cz') - \nu(hx' + by' + fz'); \end{aligned}$$

we have thus the system

$$\begin{aligned} (A, H, G | mz' - ny', \dots) &= \mu(gx' + fy' + cz') - \nu(hx' + by' + fz'), \\ (H, B, F | mz' - ny', \dots) &= \nu(ax' + hy' + gz') - \lambda(gx' + fy' + cz'), \\ (G, F, C | mz' - ny', \dots) &= \lambda(hx' + by' + fz') - \mu(ax' + hy' + gz'). \end{aligned}$$

and thence

$$\begin{aligned} &(A, \dots | mz' - ny', \dots | yz' - y'z, \dots) \\ &= \begin{vmatrix} yz' - y'z, zx' - z'x, xy' - x'y \\ ax' + hy' + gz', hx' + by' + fz', gx' + fy' + cz' \\ \lambda, \mu, \nu \end{vmatrix} \end{aligned}$$

or observing that the term in λ is

$$-(zx' - z'x)(gx' + fy' + cz') + (xy' - x'y)(hx' + by' + fz'),$$

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which is

$$\begin{aligned}
 &= x' \{ x'(ax' + hy' + gz') + y'(hx' + by' + fz') + z'(gx' + fy' + cz') \} \\
 &\quad - x \cdot x'(ax' + hy' + gz') \\
 &\quad - y \cdot x'(hx' + by' + fz') \\
 &\quad - z \cdot x'(gx' + fy' + cz') \\
 &= -x'(a, \dots | x, y, z | x', y', z') + x(a, \dots | x', y', z')^2,
 \end{aligned}$$

with similar expressions for the terms in μ, ν , we have

$$(A, \dots | mz' - ny', \dots | yz' - y'z, \dots) = -(\lambda x' + \mu y' + \nu z') \cdot (a, \dots | x, y, z | x', y', z') + (\lambda x + \mu y + \nu z) \cdot (a, \dots | x', y', z')^2,$$

$$(A, \dots | mz' - ny', \dots)^2 = -(\lambda x' + \mu y' + \nu z') \cdot (a, \dots | l, m, n | x', y', z') + (\lambda l + \mu m + \nu n) \cdot (a, \dots | x', y', z')^2,$$

where

$$(a, \dots | l, m, n | x', y', z') = \lambda x' + \mu y' + \nu z',$$

so that

$$\begin{aligned}
 \lambda l + \mu m + \nu n &= \frac{1}{K}(A, \dots | \lambda, \mu, \nu)^2, \\
 (A, \dots | mz' - ny', \dots)^2 &= -(\lambda x' + \mu y' + \nu z')^2 + \frac{1}{K}(A, \dots | \lambda, \mu, \nu)^2 \cdot (a, \dots | x', y', z')^2.
 \end{aligned}$$

Moreover,

$$\Pi = \begin{vmatrix} x, & y, & z \\ x', & y', & z' \\ l, & m, & n \end{vmatrix} = -\frac{1}{K}[(A\lambda + H\mu + G\nu)(yz' - y'z) + (H\lambda + B\mu + F\nu)(zx' - z'x) + (G\lambda + F\mu + C\nu)(xy' - x'y)],$$

which is

$$= -\frac{1}{K}(A, \dots | \lambda, \mu, \nu | yz' - y'z, \dots),$$

and hence instead of the formula I. we have

$$(a, \dots | x, y, z)^2 = \text{Quotient by } (a, \dots | x', y', z')^2 \text{ of}$$

$$\text{I. (bis) } \left\{ \begin{aligned} &[(a, \dots | x, y, z | x', y', z')]^2 \\ &+ \text{Quotient by } (A, \dots | \lambda, \mu, \nu)^2(a, \dots | x', y', z')^2 - K(\lambda x' + \mu y' + \nu z')^2 \text{ of } K \text{ into Product} \\ &\{(\lambda x' + \mu y' + \nu z') \cdot (a, \dots | x, y, z | x', y', z') - (\lambda x + \mu y + \nu z) \cdot (a, \dots | x', y', z')^2\} \\ &\pm \sqrt{-K(a, \dots | x', y', z')^2} (A, \dots | \lambda, \mu, \nu | yz' - y'z, \dots). \end{aligned} \right.$$

25. If, in like manner, in the formula II. we introduce, instead of (λ, μ, ν) , the new quantities (l, m, n) , where

$$(\lambda, \mu, \nu) = \begin{pmatrix} a, h, g \\ h, b, f \\ g, f, c \end{pmatrix} (l, m, n),$$

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or, what is the same thing,

$$(l, m, n) = \frac{1}{K} \begin{pmatrix} A, H, G \\ H, B, F \\ G, F, C \end{pmatrix} (\lambda, \mu, \nu),$$

then we have

$$\begin{aligned}
 (a, h, g | \nu\eta' - \mu\zeta', \dots) &= n(H\xi' + B\eta' + F\zeta') - m(G\xi' + F\eta' + C\zeta'), \\
 (h, b, f | \nu\eta' - \mu\zeta', \dots) &= l(G\xi' + F\eta' + C\zeta') - n(A\xi' + H\eta' + G\zeta'), \\
 (g, f, c | \nu\eta' - \mu\zeta', \dots) &= m(A\xi' + H\eta' + G\zeta') - l(H\xi' + B\eta' + F\zeta');
 \end{aligned}$$

and thence

$$(a, \dots | \nu\eta' - \mu\zeta', \dots | x, y, z) = \begin{vmatrix} & x & & & y & z \\ A\xi' + H\eta' + G\zeta', & H\xi' + B\eta' + F\zeta', & G\xi' + F\eta' + C\zeta' & & & \\ & l, & m, & n & & \end{vmatrix}$$

$$\begin{aligned}
 &= (A, \dots | mz - ny, \dots | \xi', \eta', \zeta'), \\
 (a, \dots | \nu\eta' - \mu\zeta', \dots)^2 &= \frac{1}{K} \left[(A, \dots | \lambda, \mu, \nu)^2 \cdot (A, \dots | \xi', \eta', \zeta')^2 - (A, \dots | \lambda, \mu, \nu | \xi', \eta', \zeta')^2 \right] \\
 &= (a, \dots | l, m, n)^2 \cdot (A, \dots | \xi', \eta', \zeta')^2 - K(l\xi' + m\eta' + n\zeta')^2; \\
 \Pi &= \begin{vmatrix} & x & y & z \\ A\xi' + H\eta' + G\zeta' & H\xi' + B\eta' + F\zeta' & G\xi' + F\eta' + C\zeta' & \end{vmatrix} \\
 &= (\eta z' - \zeta y')(G\xi' + F\eta' + C\zeta') - [(H\xi' + B\eta' + F\zeta')] \dots \quad [\text{bracketed determinant expansion as printed}] \\
 &= \lambda \{ (\xi'x + \eta'y + \zeta'z)(A\xi' + H\eta' + G\zeta') - x(A, \dots | \xi', \eta', \zeta')^2 \} \\
 &\quad + \mu \{ (\xi'x + \eta'y + \zeta'z)(H\xi' + B\eta' + F\zeta') - y(A, \dots | \xi', \eta', \zeta')^2 \} \\
 &\quad + \nu \{ (\xi'x + \eta'y + \zeta'z)(G\xi' + F\eta' + C\zeta') - z(A, \dots | \xi', \eta', \zeta')^2 \},
 \end{aligned}$$

and the formula II. thus becomes

$$\begin{aligned}
 (a, \dots | x, y, z)^2 &= \text{Quotient by } (A, \dots | \xi', \eta', \zeta')^2 \text{ of} \\
 \text{II. (bis)} \quad &\begin{cases} K(\xi'x + \eta'y + \zeta'z)^2 \\ + \text{Quotient by } (a, \dots | l, m, n)^2 \cdot (A, \dots | \xi', \eta', \zeta')^2 - K(l\xi' + m\eta' + n\zeta')^2 \text{ of Product} \\ \{ K(l\xi' + m\eta' + n\zeta') \cdot (\xi'x + \eta'y + \zeta'z) - K(a, \dots | x, y, z | \xi', \eta', \zeta') \cdot (a, \dots | \xi', \eta', \zeta') \} \\ \pm \sqrt{-(A, \dots | \xi', \eta', \zeta')^2} (A, \dots | mz - ny, \dots | \xi', \eta', \zeta'). \end{cases}
 \end{aligned}$$

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26. And from these we at once deduce the inverse or reciprocal formulae

$$\begin{aligned}
 (A, \dots | \xi, \eta, \zeta)^2 &= \text{Quotient by } (A, \dots | \xi', \eta', \zeta')^2 \text{ of} \\
 \text{III. (bis)} \quad &\begin{cases} [(A, \dots | \xi, \eta, \zeta | \xi', \eta', \zeta')]^2 \\ + \text{Quotient by } (a, \dots | l, m, n)^2 \cdot (A, \dots | \xi', \eta', \zeta')^2 - K(l\xi' + m\eta' + n\zeta')^2 \text{ of } K \text{ into Product} \\ \{ (l\xi' + m\eta' + n\zeta') \cdot (A, \dots | \xi, \eta, \zeta | \xi', \eta', \zeta') - (l\xi + m\eta + n\zeta) \cdot (A, \dots | \xi', \eta', \zeta')^2 \} \\ \pm \sqrt{-(A, \dots | \xi', \eta', \zeta')^2} (a, \dots | l, m, n | \eta\zeta' - \zeta\eta', \dots). \end{cases}
 \end{aligned}$$

27. And

$$\begin{aligned}
 (A, \dots | \xi, \eta, \zeta)^2 &= \text{Quotient by } (a, \dots | x', y', z')^2 \text{ of} \\
 \text{IV. (bis)} \quad &\begin{cases} K(\xi x' + \eta y' + \zeta z')^2 \\ + \text{Quotient by } (A, \dots | \lambda, \mu, \nu)^2 \cdot (a, \dots | x', y', z')^2 - K(\lambda x' + \mu y' + \nu z')^2 \text{ of Product} \\ \pm \sqrt{-K(a, \dots | x', y', z')^2} (a, \dots | x', y', z' | \mu\zeta - \nu\eta, \dots). \end{cases}
 \end{aligned}$$

These four formulae have the same geometrical significations with the original four formulae to which they correspond respectively.

28. The eight formulae become all of them the same or very similar for the quadric form $(a, \dots | x, y, z)^2 = x^2 + y^2 + z^2$, which of course implies $(A, \dots | \xi, \eta, \zeta)^2 = \xi^2 + \eta^2 + \zeta^2$. Thus selecting any one of them at pleasure, e.g. the formula II. (bis), this becomes

$$\begin{aligned}
 &\{ (x^2 + y^2 + z^2)(\xi'^2 + \eta'^2 + \zeta'^2) - (\xi'x + \eta'y + \zeta'z)^2 \} \\
 &\times \{ (l^2 + m^2 + n^2)(\xi'^2 + \eta'^2 + \zeta'^2) - (l\xi' + m\eta' + n\zeta')^2 \} \\
 &= [(l\xi' + m\eta' + n\zeta')(x\xi' + m\eta' + n\zeta') - (lx + my + nz)(\xi'^2 + \eta'^2 + \zeta'^2)]^2 \\
 &\quad + (\xi'^2 + \eta'^2 + \zeta'^2) \begin{vmatrix} \xi' & \eta' & \zeta' \\ x & y & z \\ l & m & n \end{vmatrix}^2,
 \end{aligned}$$

where the terms independent of $\xi'^2 + \eta'^2 + \zeta'^2$ destroy each other. Omitting these terms, and dividing by $\xi'^2 + \eta'^2 + \zeta'^2$, the resulting equation is found to be

$$\begin{vmatrix} \xi'^2 + \eta'^2 + \zeta'^2, & \xi'x + \eta'y + \zeta'z, & \xi'l + \eta'm + \zeta'n \\ \xi'x + \eta'y + \zeta'z, & x^2 + y^2 + z^2, & lx + my + nz \\ \xi'l + \eta'm + \zeta'n, & lx + my + nz, & l^2 + m^2 + n^2 \end{vmatrix} = 0,$$

which is a well-known identical equation.

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Article Nos. 29 to 33, relating to a single conic in connexion with an ineunt or a tangent of a conic of double contact.

29. The formulae assume a very simple form when the point of intersection of the two tangents, or the line of junction of the two ineunts of the conic, is an ineunt or a tangent of a conic having double contact with the first-mentioned conic. Thus, if to the conic

$$(a, \dots | x, y, z)^2 = 0$$

tangents are drawn from a point (x', y', z') of the conic

$$(a, \dots | x, y, z)^2 + (\xi'x + \eta'y + \zeta'z)^2 = 0,$$

then we have

$$(a, \dots | x', y', z')^2 = -(\xi'x' + \eta'y' + \zeta'z')^2;$$

and using the form I. (bis), and putting therein (ξ', η', ζ') in the place of the arbitrary quantities (λ, μ, ν) , the equation of the tangent divides out by $\xi'x' + \eta'y' + \zeta'z'$, and owing to this fact it becomes

$$\begin{aligned} & (a, \dots | x', y', z' | x, y, z) + (\xi'x' + \eta'y' + \zeta'z')(\xi'x + \eta'y + \zeta'z) \\ & \pm \frac{1}{\sqrt{K}}(A, \dots | \xi', \eta', \zeta' | yz' - y'z, zx' - z'x, xy' - x'y) = 0, \end{aligned}$$

which is of the form

$$(\alpha, \beta, \gamma | x', y', z' | x, y, z) = 0,$$

where the matrix (α, β, γ) is =

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{pmatrix} = \begin{pmatrix} a + \xi'^2, & h + \xi'\eta' + \frac{1}{\sqrt{K}}(G\xi' + F\eta' + C\zeta'), & g + \xi'\zeta' - \frac{1}{\sqrt{K}}(H\xi' + B\eta' + F\zeta') \\ h + \xi'\eta' - \frac{1}{\sqrt{K}}(G\xi' + F\eta' + C\zeta'), & b + \eta'^2, & f + \eta'\zeta' + \frac{1}{\sqrt{K}}(A\xi' + H\eta' + G\zeta') \\ g + \xi'\zeta' + \frac{1}{\sqrt{K}}(H\xi' + B\eta' + F\zeta'), & f + \eta'\zeta' - \frac{1}{\sqrt{K}}(A\xi' + H\eta' + G\zeta'), & c + \zeta'^2 \end{pmatrix}.$$

30. But instead of further developing these formulae, I prefer to consider the formulae which give the points of contact of the tangents in question, viz. the ineunts of the conic $(a, \dots | x, y, z)^2 = 0$, or the tangents through the point (x', y', z') of the conic $(a, \dots | x, y, z)^2 + (\xi'x + \eta'y + \zeta'z)^2 = 0$.

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We have as before

$$(a, \dots | x', y', z')^2 = -(\xi'x' + \eta'y' + \zeta'z')^2,$$

and using the formula IV. (bis) and writing therein (ξ', η', ζ') in the place of the arbitrary quantities (λ, μ, ν) , the equation contains the factor $\xi'x' + \eta'y' + \zeta'z'$, and dividing by this factor, and by K , the line-equation of the ineunt is

$$x'\xi + y'\eta + z'\zeta + \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z') \cdot (A, \dots | \xi', \eta', \zeta' | \xi, \eta, \zeta)$$

$$\pm \frac{1}{\sqrt{K}}(a, \dots | x', y', z' | \xi' \zeta - \zeta' \xi, \dots) = 0.$$

Selecting the positive sign, the coordinates of the corresponding ineunt are

$$\begin{aligned} x' + \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z') + \frac{1}{\sqrt{K}}\{\eta'(gx' + fy' + cz') - \zeta'(hx' + by' + fz')\}, \\ y' + \frac{1}{K}(H\xi' + B\eta' + F\zeta')(\xi'x' + \eta'y' + \zeta'z') + \frac{1}{\sqrt{K}}\{\zeta'(ax' + hy' + gz') - \xi'(gx' + fy' + cz')\}, \\ z' + \frac{1}{K}(G\xi' + F\eta' + C\zeta')(\xi'x' + \eta'y' + \zeta'z') + \frac{1}{\sqrt{K}}\{\xi'(hx' + by' + fz') - \eta'(ax' + hy' + gz')\}, \end{aligned}$$

and taking (X, Y, Z) for the coordinates of the ineunt in question, and putting for shortness

$$\begin{aligned} \alpha &= -\frac{1}{\sqrt{K}}(g\eta' - h\zeta'), & \beta &= -\frac{1}{\sqrt{K}}(f\eta' - b\zeta'), & \gamma &= -\frac{1}{\sqrt{K}}(c\eta' - f\zeta'), \\ \alpha' &= -\frac{1}{\sqrt{K}}(a\zeta' - g\xi'), & \beta' &= 1 - \frac{1}{\sqrt{K}}(h\zeta' - f\xi'), & \gamma' &= -\frac{1}{\sqrt{K}}(g\zeta' - c\xi'), \\ \alpha'' &= -\frac{1}{\sqrt{K}}(h\xi' - a\eta'), & \beta'' &= -\frac{1}{\sqrt{K}}(b\xi' - h\eta'), & \gamma'' &= 1 - \frac{1}{\sqrt{K}}(f\xi' - g\eta'), \end{aligned}$$

we may write

$$\begin{aligned} (1 + P)X &= (2 - \alpha)x' - \beta y' - \gamma z' + \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ (1 + P)Y &= -\alpha' x' + (2 - \beta')y' - \gamma' z' + \frac{1}{K}(H\xi' + B\eta' + F\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ (1 + P)Z &= -\alpha'' x' - \beta'' y' + (2 - \gamma'')z' + \frac{1}{K}(G\xi' + F\eta' + C\zeta')(\xi'x' + \eta'y' + \zeta'z'), \end{aligned}$$

where P , which is arbitrary, may be put

$$= -\frac{1}{K}(A, \dots | \xi', \eta', \zeta')^2.$$

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31. These equations then give

$$(x', y', z') = \begin{pmatrix} \alpha, \beta, \gamma \\ \alpha', \beta', \gamma' \\ \alpha'', \beta'', \gamma'' \end{pmatrix} (X, Y, Z),$$

which can be verified without difficulty by reversing the process; and we have thus the coordinates (X, Y, Z) in terms of (x', y', z') , and reciprocally.

32. If (X_1, Y_1, Z_1) are the coordinates of the other ineunt, we have, it is clear,

$$(x', y', z') = \begin{pmatrix} 2 - \alpha, -\beta, -\gamma \\ -\alpha', 2 - \beta', -\gamma' \\ -\alpha'', -\beta'', 2 - \gamma'' \end{pmatrix} (X_1, Y_1, Z_1);$$

or substituting for (x', y', z') their values in terms of (X, Y, Z) ,

$$(2X_1, 2Y_1, 2Z_1) = \begin{pmatrix} \alpha, \beta, \gamma \\ \alpha', \beta', \gamma' \\ \alpha'', \beta'', \gamma'' \end{pmatrix} (X + X_1, Y + Y_1, Z + Z_1),$$

so that $(X + X_1, Y + Y_1, Z + Z_1)$ are the same linear functions of $2X_1, 2Y_1, 2Z_1$ that (X, Y, Z) are of (x', y', z') ; that is, we have

$$\begin{aligned} \frac{1}{2}(1 + P)(X + X_1) &= (2 - \alpha)X_1 - \beta Y_1 - \gamma Z_1 + \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ \frac{1}{2}(1 + P)(Y + Y_1) &= -\alpha' X_1 + (2 - \beta')Y_1 - \gamma' Z_1 + \frac{1}{K}(H\xi' + B\eta' + F\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ \frac{1}{2}(1 + P)(Z + Z_1) &= -\alpha'' X_1 - \beta'' Y_1 + (2 - \gamma'')Z_1 + \frac{1}{K}(G\xi' + F\eta' + C\zeta')(\xi'x' + \eta'y' + \zeta'z'), \end{aligned}$$

which equations may be written

$$(1 + P)(X, Y, Z) = \begin{pmatrix} \mathfrak{a}, \mathfrak{b}, \mathfrak{c} \\ \mathfrak{a}', \mathfrak{b}', \mathfrak{c}' \\ \mathfrak{a}'', \mathfrak{b}'', \mathfrak{c}'' \end{pmatrix} (X_1, Y_1, Z_1),$$

where the values of the coefficients are

$$\begin{aligned} \mathfrak{a} &= (a\eta' - h\xi') + \frac{1}{K}\xi'(A\xi' + H\eta' + G\zeta'), & \mathfrak{b} &= -\frac{1}{\sqrt{K}}(h\eta' - b\xi') + \frac{1}{K}\xi'(H\xi' + B\eta' + F\zeta'), \\ \mathfrak{c} &= -\frac{1}{\sqrt{K}}(g\eta' - f\xi') + \frac{1}{K}\xi'(G\xi' + F\eta' + C\zeta'); & \mathfrak{a}' &= 1 - \frac{1}{\sqrt{K}}(A\xi' - h\eta') + \frac{1}{K}\eta'(A\xi' + H\eta' + G\zeta'), \\ \mathfrak{b}' &= -\frac{1}{\sqrt{K}}(B\eta' - h\zeta') + \frac{1}{K}\eta'(H\xi' + B\eta' + F\zeta'), & \mathfrak{c}' &= -\frac{1}{\sqrt{K}}(F\eta' - c\xi') + \frac{1}{K}\eta'(G\xi' + F\eta' + C\zeta'); \\ \mathfrak{a}'' &= -\frac{1}{\sqrt{K}}(g\xi' - c\eta') + \frac{1}{K}\zeta'(A\xi' + H\eta' + G\zeta'), & \mathfrak{b}'' &= -\frac{1}{\sqrt{K}}(b\xi' - h\zeta') + \frac{1}{K}\zeta'(H\xi' + B\eta' + F\zeta'), \\ \mathfrak{c}'' &= 1 - \frac{1}{\sqrt{K}}(f\xi' - g\eta') + \frac{1}{K}\zeta'(G\xi' + F\eta' + C\zeta'). \end{aligned}$$

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and considering (X, Y, Z) and (X_1, Y_1, Z_1) as quantities connected by the foregoing linear relations, we have identically

$$(a, \dots | X, Y, Z)^2 = (a, \dots | X_1, Y_1, Z_1)^2.$$

The investigation leads thus to the automorphic transformation of the quadric function, a transformation first effected by M. Hermite².

33. It is to be remarked that the foregoing formulae show that (x', y', z') being the coordinates of a point on the conic $(a, \dots | x, y, z)^2 + (\xi'x + \eta'y + \zeta'z)^2 = 0$, from which point tangents are drawn to the conic $(a, \dots | x, y, z)^2 = 0$, the ineunts (or points of contact), enter linearly into the equations of the tangents, the ineunts (or coordinates of contact), and the polar. And it may be added that the equation of the conic enveloped by the polar (that is, the polar conic of $(a, \dots | x, y, z)^2 + (\xi'x + \eta'y + \zeta'z)^2 = 0$) has for its equation

$$\{K + (A, \dots | \xi', \eta', \zeta')^2\}(a, \dots | x, y, z)^2 - K(\xi'x + \eta'y + \zeta'z)^2 = 0,$$

and that the coordinates of the point of contact of the polar with this conic are

$$\begin{aligned} x' &+ \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ y' &+ \frac{1}{K}(H\xi' + B\eta' + F\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ z' &+ \frac{1}{K}(G\xi' + F\eta' + C\zeta')(\xi'x' + \eta'y' + \zeta'z'); \end{aligned}$$

so that (x', y', z') also enter linearly into the expressions for the coordinates of the last-mentioned point.

Article Nos. 34 to 37, relating to two conics.

34. Considering now the two conics

$$U = (a, b, c, f, g, h | x, y, z)^2 = 0,$$

$$U' = (a', b', c', f', g', h' | x, y, z)^2 = 0;$$

suppose that the conic

$$\theta U + \theta' U' = (\theta a + \theta' a', \dots | x, y, z)^2 = 0$$

represents a pair of lines.

The condition for this is

$$\text{Disc.}(\theta a + \theta' a', \dots | x, y, z)^2 = 0,$$

²See my "Memoir on the Automorphic Transformation of a Bipartite Quadric Function," *Phil. Trans.*, t. cxlviii. (1858), pp. 39–46, [115].

which is

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D} \mid \theta, \theta')^3 = 0,$$

where

$$\begin{aligned}\mathfrak{A} &= K, & \mathfrak{D} &= K', \\ \mathfrak{B} &= Aa' + Bb' + Cc' + 2Ff' + 2Gg' + 2Hh', \\ \mathfrak{C} &= A'a + B'b + C'c + 2F'f + 2G'g + 2H'h,\end{aligned}$$

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(the significations of $K', A', B', C', F', G', H'$ being of course analogous to those of K, A, B, C, F, G, H). The three roots $\theta : \theta'$ correspond, it is clear, to the three pairs of lines which can be drawn through the intersections of the two conics.

35. The equation

$$\text{Disc.}(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D} \mid \theta, \theta')^3 = 0,$$

which is of the fourth order in $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$, and of the sixth order as regards (a, b, c, f, g, h) and (a', b', c', f', g', h') respectively, is the condition, in order that the two conics may touch each other. Assuming that it is satisfied, the cubic equation in $\theta : \theta'$ has a pair of equal roots; or say there is a twofold root and a onefold root; the twofold root gives the pair of lines drawn from the point of contact to the other two points of intersection, the onefold root gives the pair made up of the common tangent and the line joining the other two points of intersection.

36. In particular, suppose that the two conics are

$$2(\rho x + \sigma y + \tau z)(\rho' x + \sigma' y + \tau' z) = 0,$$

$$2(\lambda x + \mu y + \nu z)(\lambda' x + \mu' y + \nu' z) = 0;$$

so that

$$\begin{aligned}(a, b, c, f, g, h) &= (2\rho\rho', 2\sigma\sigma', 2\tau\tau', \sigma\tau' + \sigma'\tau, \tau\rho' + \tau'\rho, \rho\sigma' + \rho'\sigma), \\ (a', b', c', f', g', h') &= (2\lambda\lambda', 2\mu\mu', 2\nu\nu', \mu\nu' + \mu'\nu, \nu\lambda' + \nu'\lambda, \lambda\mu' + \lambda'\mu), \\ (A, B, C, F, G, H) &= -(\sigma\tau' - \sigma'\tau, \tau\rho' - \tau'\rho, \rho\sigma' - \rho'\sigma, \dots)^2, \\ (A', B', C', F', G', H') &= -(\mu\nu' - \mu'\nu, \nu\lambda' - \nu'\lambda, \lambda\mu' - \lambda'\mu, \dots)^2;\end{aligned}$$

and thence also

$$\begin{aligned}\mathfrak{A} &= K = 0, & \mathfrak{D} &= K' = 0, \\ \mathfrak{B} &= Aa' + \&c. = -2 \begin{vmatrix} \lambda, \mu, \nu \\ \lambda', \mu', \nu' \\ \sigma\tau', \tau\rho', \rho\sigma' \end{vmatrix} \cdot \begin{vmatrix} \lambda, \mu, \nu \\ \lambda', \mu', \nu' \\ \sigma'\tau, \tau'\rho, \rho'\sigma \end{vmatrix}, \\ \mathfrak{C} &= A'a + \&c. = -2 \begin{vmatrix} \rho, \sigma, \tau \\ \rho', \sigma', \tau' \\ \mu\nu', \nu\lambda', \lambda\mu' \end{vmatrix} \cdot \begin{vmatrix} \rho, \sigma, \tau \\ \rho', \sigma', \tau' \\ \mu'\nu, \nu'\lambda, \lambda'\mu \end{vmatrix};\end{aligned}$$

and the equation in (θ, θ') is

$$\mathfrak{B}\theta^2 + \mathfrak{C}\theta'^2 = 0;$$

hence writing $\theta = \mathfrak{C}$, $\theta' = -\mathfrak{B}$, the equation of the pair of lines is

$$\begin{vmatrix} \rho, \sigma, \tau \\ \rho', \sigma', \tau' \\ \mu\nu', \nu\lambda', \lambda\mu' \end{vmatrix} \begin{vmatrix} \rho, \sigma, \tau \\ \rho', \sigma', \tau' \\ \mu'\nu, \nu'\lambda, \lambda'\mu \end{vmatrix} (\rho x + \sigma y + \tau z)(\rho' x + \sigma' y + \tau' z) - \begin{vmatrix} \lambda, \mu, \nu \\ \lambda', \mu', \nu' \\ \sigma\tau', \tau\rho', \rho\sigma' \end{vmatrix} \begin{vmatrix} \lambda, \mu, \nu \\ \lambda', \mu', \nu' \\ \sigma'\tau, \tau'\rho, \rho'\sigma \end{vmatrix} (\lambda x + \mu y + \nu z)(\lambda' x + \mu' y + \nu' z) = 0;$$

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and it is easy to see that the left-hand side does in fact break up into factors, and that the equation is

$$\left| \begin{array}{ccc} x, y, z \\ \mu\tau' - \nu\sigma', \nu\rho' - \lambda\tau', \lambda\sigma' - \mu\rho' \\ \sigma'\nu - \tau'\mu, \tau'\lambda - \rho'\nu, \rho'\mu - \sigma'\lambda \end{array} \right| \cdot \left| \begin{array}{ccc} x, y, z \\ \mu'\tau - \nu'\sigma, \nu'\rho - \lambda'\tau, \lambda'\sigma - \mu'\rho \\ \sigma\nu' - \tau\mu', \tau\lambda' - \rho\nu', \rho\mu' - \sigma\lambda' \end{array} \right| = 0,$$

which of course might have been obtained at once by means of the four points which are the intersection of each component line of the first conic by each component line of the second conic.

37. Suppose that the first conic is

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0,$$

while the second conic is the pair of lines

$$2(\lambda x + \mu y + \nu z)(\lambda'x + \mu'y + \nu'z) = 0;$$

then, putting as before,

$$(\theta a + \theta' \cdot 2\lambda\lambda', \dots \mid x, y, z)^2 = 0,$$

we have

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D} \mid \theta, \theta')^3 = 0,$$

where

$$\mathfrak{A} = K,$$

$$\mathfrak{B} = 2(A, B, C, F, G, H \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu'),$$

$$\mathfrak{C} = -(a, b, c, f, g, h \mid \mu\nu' - \mu'\nu, \nu\lambda' - \nu'\lambda, \lambda\mu' - \lambda'\mu)^2,$$

$$\mathfrak{D} = 0;$$

and the equation in (θ, θ') is

$$K\theta^3 + 2(A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu')\theta^2\theta' - (a, \dots \mid \mu\nu' - \mu'\nu, \dots)^2\theta\theta'^2 = 0,$$

which may be written

$$[K\theta + (A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu')\theta']^2 = \left[[(A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu')]^2 + K(a, \dots \mid \mu\nu' - \mu'\nu, \dots)^2 \right] \theta'^2,$$

that is

$$= (A, \dots \mid \lambda, \mu, \nu)^2 \cdot (A, \dots \mid \lambda', \mu', \nu')^2 \cdot \theta'^2,$$

$$K\theta = [\pm\sqrt{(A, \dots \mid \lambda, \mu, \nu)^2} \sqrt{(A, \dots \mid \lambda', \mu', \nu')^2} - (A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu')]\theta';$$

we may assume

$$\theta = \pm\sqrt{(A, \dots \mid \lambda, \mu, \nu)^2} \sqrt{(A, \dots \mid \lambda', \mu', \nu')^2} - (A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu'), \quad \theta' = K,$$

so that the conic

$$\begin{aligned} & \{ \pm\sqrt{(A, \dots \mid \lambda, \mu, \nu)^2} \sqrt{(A, \dots \mid \lambda', \mu', \nu')^2} - (A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu') \} (a, \dots \mid x, y, z)^2 \\ & + 2K(\lambda x + \mu y + \nu z)(\lambda'x + \mu'y + \nu'z) = 0 \end{aligned}$$

breaks up into a pair of lines.

Putting for shortness

$$\pm\sqrt{(A, \dots \mid \lambda, \mu, \nu)^2} \sqrt{(A, \dots \mid \lambda', \mu', \nu')^2} - (A, \dots \mid \lambda, \mu, \nu \mid \lambda', \mu', \nu') = \Omega,$$

the coefficients on the left-hand side of the equation are

$$(\Omega a + 2K\lambda\lambda', \dots, \Omega f + K(\mu\nu' + \mu'\nu), \dots),$$

whence, after all reductions, the inverse function is

$$\{(A, \dots | \lambda, \mu, \nu | \xi, \eta, \zeta) \sqrt{(A, \dots | \lambda', \mu', \nu')^2} + (A, \dots | \lambda', \mu', \nu' | \xi, \eta, \zeta) \sqrt{(A, \dots | \lambda, \mu, \nu)^2}\}^2,$$

and the remainder of the process of decomposition is effected without difficulty.

Addition, 18 December, 1862.

The formulae II. and II. (bis) each of them give the tangents of the conic $(a, \dots | x, y, z)^2 = 0$ at the ineunts of intersection with the line $\xi'x + \eta'y + \zeta'z = 0$. A very elegant formula for these ineunts themselves was communicated to me by Mr Spottiswoode, and I have since found that the same or an equivalent formula is made use of by M. Aronhold in his recent valuable memoir, “Ueber eine neue algebraische Behandlungsweise der Integrale irrationaler Differentiale, &c.,” *Crelle*, t. LXI. pp. 95–145 (1862). The formula is as follows, viz. for the conic and line,

$$(a, b, c, f, g, h | x, y, z)^2 = 0,$$

$$\xi'x + \eta'y + \zeta'z = 0,$$

then

$$\begin{aligned} x : y : z = \\ (l\xi' + m\eta' + n\zeta') \frac{d\phi}{d\xi'} + \eta'(gl + fm + cn) - \zeta'(hl + bm + fn) + l\sqrt{\phi}, \\ : (l\xi' + m\eta' + n\zeta') \frac{d\phi}{d\eta'} + \zeta'(al + hm + gn) - \xi'(gl + fm + cn) + m\sqrt{\phi}, \\ : (l\xi' + m\eta' + n\zeta') \frac{d\phi}{d\zeta'} + \xi'(hl + bm + fn) - \eta'(al + hm + gn) + n\sqrt{\phi}; \end{aligned}$$

where

$$\phi = \begin{vmatrix} \xi' & \eta' & \zeta' \\ \xi' & a & h & g \\ \eta' & h & b & f \\ \zeta' & g & f & c \end{vmatrix} = -(A, B, C, F, G, H | \xi', \eta', \zeta')^2,$$

so that $\frac{1}{2} \frac{d\phi}{d\xi'}$, $\frac{1}{2} \frac{d\phi}{d\eta'}$, $\frac{1}{2} \frac{d\phi}{d\zeta'}$ are respectively

$$= -(A\xi' + H\eta' + G\zeta'), \quad -(H\xi' + B\eta' + F\zeta'), \quad -(G\xi' + F\eta' + C\zeta'),$$

and where l, m, n are supernumerary arbitrary quantities.

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ON THE WAVE SURFACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 16–22.]

SOME very beautiful results in relation to the Wave Surface have been recently obtained by Herr Zech, in the Memoirs “Die Eigenschaften der Wellenflächen der zweiaxigen Krystalle mittels der höhern Geometrie abgeleitet,” *Crelle*, t. LII. pp. 243–254 (1856), and “Die Krümmungslinien der Wellenfläche zweiaxiger Krystalle,” *Crelle*, t. LIV. pp. 72–77 (1857). According to the former of Fresnel’s two modes of generation, the wave surface is the envelope of a plane whose perpendicular distance v is a certain given function of the direction cosines l, m, n . For the same system of direction cosines, there are in fact two values of the perpendicular distance: call these v and w , and let the corresponding planes (parallel of course to each other) be called P and Q . Then the entire system of the planes P and Q envelope the wave surface, viz. the planes P may be considered as enveloping one sheet and the planes Q the other sheet of the surface.

But if instead of considering the entire system of planes we consider only the planes P and the parallel planes Q , for which the perpendicular on the plane P has a given constant value v , then the planes P will envelope a developable F , and the planes Q will envelope a developable G , these two developables being, it is to be observed, distinct surfaces, not sheets of one and the same surface. Or, what is the same thing, the planes P for which the perpendicular distance has a given constant value v will envelope a developable F , and the planes P for which the perpendicular distance of the parallel planes Q has a given constant value w , will envelope a developable G . And it is obvious that the developables F and G touch the wave surface along curves. The equation of the developable F contains of course the arbitrary parameter v , and the equation of the developable G contains in like manner the arbitrary parameter w , so that we in fact have two series of developables

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F and G respectively touching the wave surface along two series of curves. And it is shown in the second of the Memoirs above referred to that these curves are the curves of curvature of the wave surface. The developable F is obtained as the envelope of the plane P , whose equation is

$$lx + my + nz = v,$$

where v has a given constant value and l, m, n are parameters which vary, subject to the two conditions

$$l^2 + m^2 + n^2 = 1,$$

$$\frac{l^2}{a^2 - v^2} + \frac{m^2}{b^2 - v^2} + \frac{n^2}{c^2 - v^2} = 0.$$

And in like manner the developable G is obtained as the envelope of the plane Q , whose equation is

$$lx + my + nz = \frac{abc}{w} \sqrt{\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2}},$$

where w has a given constant value and l, m, n are parameters which vary, subject to the two conditions

$$l^2 + m^2 + n^2 = 1,$$

$$\frac{l^2}{a^2 - w^2} + \frac{m^2}{b^2 - w^2} + \frac{n^2}{c^2 - w^2} = 0.$$

{It is hardly necessary to remark that if in the last-mentioned system of equations, the parameter w is also treated as variable, we obtain the wave surface: in fact v^2, w^2 being the roots of the equation

$$\frac{l^2}{a^2 - \theta} + \frac{m^2}{b^2 - \theta} + \frac{n^2}{c^2 - \theta} = 0,$$

we have, attending to the condition $l^2 + m^2 + n^2 = 1$, the identical equation

$$l^2(b^2 - \theta)(c^2 - \theta) + m^2(c^2 - \theta)(a^2 - \theta) + n^2(a^2 - \theta)(b^2 - \theta) = (v^2 - \theta)(w^2 - \theta),$$

and thence

$$vw = \frac{abc}{w} \sqrt{\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2}},$$

which shows that the system of equations for the plane Q (w being treated as variable) is in fact identical with the system of equations for the plane P .}

The form of the equations of the planes P and Q respectively shows that each of these planes is parallel to a tangent plane of the cone

$$\frac{l^2}{a^2 - v^2} + \frac{m^2}{b^2 - v^2} + \frac{n^2}{c^2 - v^2} = 0.$$

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or in other words, that the planes P and Q are respectively tangents to the conic or infinitely thin surface of the second order, which is the section of the last-mentioned cone by the plane at infinity. Moreover it is obvious from the same equations that the plane P is a tangent plane of the sphere whose equation is

$$x^2 + y^2 + z^2 = v^2,$$

and that the plane Q is a tangent plane of the ellipsoid whose equation is

$$a^2x^2 + b^2y^2 + c^2z^2 = \frac{a^2b^2c^2}{w^2}.$$

Hence each of the developables F and G is the envelope of a plane which is the common tangent plane of two surfaces of the second order; such developables are in general of the eighth order, see my paper “On the Developable Surfaces which arise from Two Surfaces of the Second Order,” *Camb. and Dubl. Math. Journ.*, t. v. pp. 46–57 (1850), [84], and it will be presently seen that this is in fact the order of the developables F and G respectively.

The before-mentioned cone

$$\frac{l^2}{a^2 - v^2} + \frac{m^2}{b^2 - v^2} + \frac{n^2}{c^2 - v^2} = 0,$$

(Zech’s cone K) is a cone having for its focal lines the optic axes (or normals to the circular sections) of the ellipsoid $a^2x^2 + b^2y^2 + c^2z^2 = 1$ (Zech’s ellipsoid E) which is used in the theory of the Wave Surface in the place of Fresnel’s Surface of Elasticity. The complementary cone

$$(a^2 - v^2)x^2 + (b^2 - v^2)y^2 + (c^2 - v^2)z^2 = 0,$$

(Zech’s cone C) meets the last-mentioned ellipsoid in a curve lying on the sphere whose equation is $x^2 + y^2 + z^2 = \frac{1}{v^2}$, a property which may be considered as affording the geometrical construction of the magnitude v , by means of the cone K . The ellipsoid $a^2x^2 + b^2y^2 + c^2z^2 = \frac{a^2b^2c^2}{w^2}$ is obviously an ellipsoid similar to the ellipsoid E , and the value of v being determined as above, the sphere $x^2 + y^2 + z^2 = \frac{1}{v^2}$ and the ellipsoid $a^2x^2 + b^2y^2 + c^2z^2 = \frac{a^2b^2c^2}{w^2}$ (which are used in the preceding geometrical construction of the developables F and G respectively) may be considered as given by construction.

To find the equation of the developable F , we have, from the equations

$$lx + my + nz = v,$$

$$l^2 + m^2 + n^2 = 1,$$

$$\frac{l^2}{a^2 - v^2} + \frac{m^2}{b^2 - v^2} + \frac{n^2}{c^2 - v^2} = 0,$$

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treating them by the method of arbitrary multipliers in the usual manner, we obtain

$$x + \left(\rho + \frac{\theta}{a^2 - v^2} \right) l = 0,$$

$$y + \left(\rho + \frac{\theta}{b^2 - v^2} \right) m = 0,$$

$$z + \left(\rho + \frac{\theta}{c^2 - v^2} \right) n = 0,$$

equations which give in the first instance $v + \rho = 0$ or $\rho = -v$, and then substituting this value for ρ ,

$$l = \frac{x(v^2 - a^2)}{v(v^2 - a^2) + \theta}, \quad m = \frac{y(v^2 - b^2)}{v(v^2 - b^2) + \theta}, \quad n = \frac{z(v^2 - c^2)}{v(v^2 - c^2) + \theta},$$

and thence

$$\frac{x^2(v^2 - a^2)}{v(v^2 - a^2) + \theta} + \frac{y^2(v^2 - b^2)}{v(v^2 - b^2) + \theta} + \frac{z^2(v^2 - c^2)}{v(v^2 - c^2) + \theta} = v,$$

$$\frac{x^2(v^2 - a^2)}{[v(v^2 - a^2) + \theta]^2} + \frac{y^2(v^2 - b^2)}{[v(v^2 - b^2) + \theta]^2} + \frac{z^2(v^2 - c^2)}{[v(v^2 - c^2) + \theta]^2} = 0,$$

the latter of which is the derived equation of the former with respect to the parameter θ , hence writing the former equation under the form

$$\frac{x^2(v^2 - a^2)}{\theta + v(v^2 - a^2)} + \frac{y^2(v^2 - b^2)}{\theta + v(v^2 - b^2)} + \frac{z^2(v^2 - c^2)}{\theta + v(v^2 - c^2)} = v,$$

or what is the same thing,

$$(A, B, C, D \mid \theta, 1)^3 = 0,$$

where

$$A = 3v^2,$$

$$B = (v^2 - x^2)(v^2 - a^2) + (v^2 - y^2)(v^2 - b^2) + (v^2 - z^2)(v^2 - c^2),$$

$$C = (v^2 - y^2 - z^2)(v^2 - b^2)(v^2 - c^2) + (v^2 - z^2 - x^2)(v^2 - c^2)(v^2 - a^2) + (v^2 - x^2 - y^2)(v^2 - a^2)(v^2 - b^2),$$

the equation of the developable F is

$$(AD - BC)^2 - 4(AC - B^2)(BD - C^2) = 0.$$

The investigation for the developable G is very similar to the preceding. Write for shortness

$$\frac{abc}{w} \sqrt{\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2}} = \Lambda,$$

then the system of equations is

$$lx + my + nz - \Lambda = 0,$$

$$l^2 + m^2 + n^2 = 1,$$

$$\frac{l^2}{a^2 - w^2} + \frac{m^2}{b^2 - w^2} + \frac{n^2}{c^2 - w^2} = 0,$$

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and we then have

$$x - \frac{abc}{wa^2\Lambda} l + \left(\rho + \frac{\theta}{a^2 - w^2} \right) l = 0,$$

$$y - \frac{abc}{wb^2\Lambda} m + \left(\rho + \frac{\theta}{b^2 - w^2} \right) m = 0,$$

$$z - \frac{abc}{wc^2\Lambda} n + \left(\rho + \frac{\theta}{c^2 - w^2} \right) n = 0,$$

equations which give $\rho = 0$, and substituting this value, we obtain

$$l = \frac{wa^2\Lambda x}{abc - wa^2\Lambda \frac{\theta}{a^2 - w^2}} \quad \dots$$

Substituting these values in the equations

$$\frac{l^2}{a^2 - w^2} + \frac{m^2}{b^2 - w^2} + \frac{n^2}{c^2 - w^2} = 0,$$

$$\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2} = \Lambda^2,$$

we find

$$\frac{x^2}{\left\{ \frac{abc}{wa^2\Lambda} (a^2 - w^2) - \theta \right\}^2} (a^2 - w^2) + \dots = 0,$$

$$\frac{x^2\left(\frac{abc}{wa^2} - \theta\right)^2}{\left\{\frac{abc}{wa^2\Lambda}(a^2 - w^2) - \theta\right\}^2} + \dots = \Lambda^2,$$

and multiplying the first equation by $-\theta$ and the second equation by Λ^2 and adding, we obtain

$$\frac{x^2(w^2 - a^2)}{\frac{abc}{wa^2}(a^2 - w^2) - \theta} + \frac{y^2(w^2 - b^2)}{\frac{abc}{wb^2}(b^2 - w^2) - \theta} + \frac{z^2(w^2 - c^2)}{\frac{abc}{wc^2}(c^2 - w^2) - \theta} = \frac{abc}{w},$$

of which equation the latter of the foregoing two equations is the derived equation with respect to the parameter Λ^2 .

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Comparing this with the foregoing equation

$$\frac{x^2(v^2 - a^2)}{v(v^2 - a^2) + \theta} + \frac{y^2(v^2 - b^2)}{v(v^2 - b^2) + \theta} + \frac{z^2(v^2 - c^2)}{v(v^2 - c^2) + \theta} = v,$$

we see that it is deduced from it by writing

$$a^2x^2, b^2y^2, c^2z^2, \frac{w^2}{a^2} - 1, \frac{w^2}{b^2} - 1, \frac{w^2}{c^2} - 1, \frac{abc}{w}, \Lambda\theta$$

in the place of

$$x^2, y^2, z^2, v^2 - a^2, v^2 - b^2, v^2 - c^2, v, \theta$$

respectively, and the equation of the developable G is therefore

$$(A'D' - B'C')^2 - 4(A'C' - B'^2)(B'D' - C'^2) = 0,$$

where A', B', C', D' are what A, B, C, D become when

$$x^2, y^2, z^2, v^2 - a^2, v^2 - b^2, v^2 - c^2, v, \theta$$

are replaced by

$$a^2x^2, b^2y^2, c^2z^2, \frac{w^2}{a^2} - 1, \frac{w^2}{b^2} - 1, \frac{w^2}{c^2} - 1, \frac{abc}{w}, \Lambda\theta.$$

The equations of the developables F and G , although radically distinct from each other, are consequently similar in form, and each is at once deducible from the other.

H

2, Stone Buildings, W.C., 9th March, 1858.

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NOTE ON THE SINGULAR SOLUTIONS OF DIFFERENTIAL EQUATIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 36, 37.]

THE following investigation (which has been in my possession for a good many years) affords I think a simple explanation of the theory of the singular solutions of differential equations.

Let the primitive equation be

$$c^n + Pc^{n-1} + Qc^{n-2} + \dots = 0,$$

where c is the arbitrary constant and $P, Q \dots$ are any functions of x, y ; then the differential equation is obtained by eliminating c from the foregoing equation and the derived equation

$$P'c^{n-1} + Q'c^{n-2} + \dots = 0,$$

and the result may be represented by

$$F(P, Q, \dots, P', Q', \dots) = 0.$$

Assume now

$$c^n + Pc^{n-1} + Qc^{n-2} + \dots = (c + X)(c + Y)(c + Z) \dots,$$

then we have

$$\begin{aligned} P \text{ \&c.} &= X + Y + Z + \text{\&c.}, \\ Q &= XY + XZ + YZ + \text{\&c.}, \end{aligned}$$

and consequently

$$\begin{aligned} P' \text{ \&c.} &= X' + Y' + Z' + \text{\&c.}, \\ Q' &= (Y + Z + \text{\&c.})X' + \text{\&c.}, \end{aligned}$$

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and substituting these values in the function $F(P, Q, \dots, P', Q', \dots)$ it is clear that we shall have $F(P, Q, \dots, P', Q', \dots) = UX'Y'Z' \dots$ where U is a symmetrical function of $X, Y, Z, \text{\&c.}$, and therefore a function of $P, Q, \dots$; and this equation will be identically true whatever values we attribute to $X', Y', Z', \dots$, hence putting these quantities respectively equal to unity, we have

$$\begin{aligned} P' &= n, \\ Q' &= (n - 1)P, \\ R' \text{ \&c.} &= (n - 2)Q, \end{aligned}$$

and with these values

$$U = F(P, Q, \dots, P', Q', \dots),$$

that is $U = 0$ is the result obtained by eliminating c from the primitive equation and the equation

$$nc^{n-1} + (n - 1)Pc^{n-2} + \dots = 0,$$

which is the equation obtained by differentiating the primitive equation with respect to the arbitrary constant c : that is, $U = 0$ being the singular solution, the differential equation is

$$UX'Y'Z' \dots = 0.$$

It is to be remarked that ($P, Q, \text{\&c.}$ being rational and integral functions) then if the roots $X, Y, Z, \text{\&c.}$ are also rational and integral functions, the differential equation contains the differential equation does not really contain the rational and integral factor U , but $X'Y'Z' \dots$ is here a rational fraction containing U in the denominator and $UX'Y'Z' \dots$ is an indecomposable rational and integral function. This is easily verified *à posteriori* for a quadratic equation.

2, Stone Buildings, W.C., 28th January, 1858.

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ON A THEOREM RELATING TO SPHERICAL CONICS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), p. 53.]

THE following theorem was given by Prof. MacCullagh: “If three lines at right angles to each other pass through a fixed point O so that two of them are confined to given planes: the third line traces out a cone of the second order whose sections parallel to the given planes are circles, and the plane containing

the other two lines envelopes a cone of the second order whose sections by planes parallel to the given planes are parabolas.”

Referring the figure to the sphere we have a trirectangular triangle XYZ , of which two angles X, Y lie on fixed arcs A, B . The angle Z generates a spherical conic U' having A, B for its cyclic arcs. The side XY envelopes a spherical conic U touched by the arcs A, B . The conic U' is evidently the supplementary conic of U , hence the poles of A, B are the foci of U . We may drop altogether the consideration of the triangle XYZ and consider only the side XY , we have then the theorem:

If a quadrantal arc XY slides between the two fixed arcs A, B , the envelope of XY is a spherical conic U touched by the fixed arcs A, B , and which has for its foci the poles of these same arcs A, B .

It is worth while to notice the great reduction of order which takes place in consequence of the arc XY being a quadrant. If XY had been an arc of a given magnitude θ , the envelope would have been a spherical curve of an order certainly higher than 6. For considering the corresponding problem *in plano*, the envelope in the particular case where the fixed lines A, B are at right angles to each other is a curve of the sixth order, and in the general case where the two fixed lines are not at right angles the order is higher: the problem *in plano* corresponds of course, not to the general problem on the sphere, but to that in which θ is indefinitely small.

[continued p. 429]